

NADEN MOHAMED YASEN

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SUMMARY

AI & Computer Science graduate (Helwan University, 2026) with hands-on experience developing ML, NLP, and GenAI systems, from model development through Docker-containerized, FastAPI-served deployment with Prometheus/Grafana monitoring. Practical experience with RAG, LLM fine-tuning (LoRA/PEFT), and offline evaluation harnesses, paired with a habit of independently debugging issues through systematic, log-driven analysis.

EDUCATION

B.Sc. Computer Science & Artificial Intelligence · Helwan University, Cairo

Sep 2022 – Jun 2026

GPA: 3.37/4.0 (Very Good with honors)

SKILLS

Programming Fundamentals: Python, C++, Object-Oriented Programming (OOP), Data Structures, Algorithms, SOLID Principles, Design Patterns

Web & Backend Development: FastAPI, REST API Design, HTML, CSS, Postman

Databases & Data Storage: SQL, PostgreSQL, pgvector, SQLAlchemy, Alembic, MongoDB Atlas, Qdrant, FAISS, SQLite

Machine Learning & Statistics: Statistics, Probability, Linear Algebra, Calculus, Scikit-learn, Model Evaluation & Benchmarking, EDA, Data Cleaning, Feature Engineering, Data Preprocessing, Predictive Modeling

Deep Learning & Generative AI: PyTorch, LLMs, RAG, Fine-Tuning (LoRA/PEFT), NLP, Agentic Systems & Tool-Calling, Hugging Face, LangChain, LangGraph

MLOps & DevOps: Docker, Docker Compose (multi-service orchestration), Prometheus, Grafana, Nginx, database migrations, API monitoring & observability, Structured logging, Weights & Biases (W&B)

Testing & Developer Tools: PyTest, Git/GitHub, Jupyter Notebook, Google Colab, VS Code, Streamlit, Gradio

Languages Spoken: Arabic, English

RELEVANT PROJECTS

ScholarFlow AI – RAG-based Document Intelligence Platform | *Python, FastAPI, PostgreSQL/pgvector, Qdrant, Docker*

- Designed and built a RAG backend with a provider-factory architecture, allowing the LLM (Groq) and vector database (pgvector/Qdrant) to be swapped via configuration with no code changes.
- Designed the PostgreSQL schema and managed migrations (Alembic) for projects, uploaded assets, and vector-indexed chunks.
- Containerized the full system with Docker Compose across 7 services (FastAPI, PostgreSQL/pgvector, Qdrant, Nginx, Prometheus, Grafana, exporters), including a custom migration-then-serve entry point workflow.
- Instrumented the API with Prometheus metrics (latency, throughput) and Grafana dashboards, then extended the stack with RAG-specific metrics (empty-retrieval rate, retrieval confidence distribution) to monitor answer quality in real time, not just uptime.
- Built an offline evaluation harness combining deterministic retrieval metrics (hit rate, MRR) with RAGAS-based generation metrics (faithfulness, answer relevancy) to catch quality regressions before deployment.

ELARA – Clinical-Grade RAG System for Maternal & Neonatal Guidance | *AI Hackathon 2026, Orange Digital Center (Orange Egypt) & Instant Software Solutions · FastAPI, MongoDB Atlas, Qdrant, Docling, BAAI/bge-m3, pytest*

- Built the backend and AI architecture for a citation-grounded RAG system translating clinical guidelines into safe, trustworthy patient guidance, in a 5-day hackathon sprint.
- Designed a structure-aware ingestion pipeline (Docling) and hybrid dense + sparse retrieval on Qdrant with multilingual BAAI/bge-m3 embeddings for accurate Arabic/English medical-term retrieval.
- Engineered a deterministic safety layer — pre-generation risk/prompt-injection filtering and post-generation citation verification — to ground every response and eliminate hallucination.
- Built an evaluation harness (precision@k, recall@k, MRR, faithfulness, answer relevancy) backed by a unit/integration/E2E pytest suite.

Adaptive Code Intelligence Assistant | *LLM Evaluation, Pydantic, pytest*

- Designed a multi-step LLM pipeline with strict Pydantic schemas at every stage, keeping component handoffs reliable and testable.
- Developed an evaluation harness to measure router classification accuracy and end-to-end code-generation correctness (pass@1) against a benchmark dataset, transforming LLM out

Toxic Content Classification System | *PyTorch, DistilBERT, LoRA, Llama Guard, BLIP*

- Evaluated and compared modeling approaches (LSTM baseline vs. LoRA-fine-tuned DistilBERT), tracking accuracy, precision, recall, and F1 across categories using Weights & Biases.
- Tested a zero-shot content moderation pipeline (Llama Guard) alongside a multimodal BLIP captioning module, weighing trade-offs between approaches ahead of Streamlit deployment.

TRAINING & CERTIFICATIONS

IBM-Certified AI & Data Science Professional | Digital Egypt Pioneers Initiative (DEPI) Nov 2024 – May 2025

- Completed an IBM-certified program covering Python, ML, SQL, and MLOps; built and evaluated ML/DL models, achieving 91.12% accuracy on the capstone project.

NLP Training | Neurova AI Apr 2026 – Jul 2026

- Completed an intensive 8-week NLP training program spanning classical preprocessing, TF-IDF, Transformer architectures, BERT fine-tuning, RAG, AI Agents and LLM integration across 7 end-to-end projects.

AI Hackathon 2026 | Orange Digital Center (Orange Egypt) & Instant Software Solutions Aug 2026

- Collaborated with a 4-person team to build ELARA, a citation-grounded RAG system translating clinical guidelines into safe, trustworthy maternal & neonatal guidance with safety and validation layers, in a 5-day sprint, finishing among the top 7 teams

Data Analysis Hackathon | Orange Digital Center (Orange Egypt) & Instant Software Solutions Feb 2026

- Developed an end-to-end data analysis pipeline scraping 30,000+ European hotel listings from Booking.com using Python and Selenium, designed a relational database schema with ERD and SQL analytics, and delivered interactive business intelligence dashboards in Streamlit and Power BI to answer strategic pricing and guest satisfaction questions across major European markets.